

- 4 The successive ionisation energies (IE) of two elements, **W** and **X**, are given below.

| IE / kJ mol <sup>-1</sup> | 1 <sup>st</sup> | 2 <sup>nd</sup> | 3 <sup>rd</sup> | 4 <sup>th</sup> | 5 <sup>th</sup> | 6 <sup>th</sup> | 7 <sup>th</sup> | 8 <sup>th</sup> |
|---------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| <b>W</b>                  | 1090            | 2350            | 4610            | 6220            | 37800           | 47000           | -               | -               |
| <b>X</b>                  | 1251            | 2298            | 3822            | 5158            | 6542            | 9362            | 11018           | 33604           |

What is the likely formula of the compound that is formed when **W** reacts with **X**?

- A**    **WX**                      **B**    **W<sub>2</sub>X<sub>3</sub>**                      **C**    **WX<sub>4</sub>**                      **D**    **W<sub>4</sub>X**